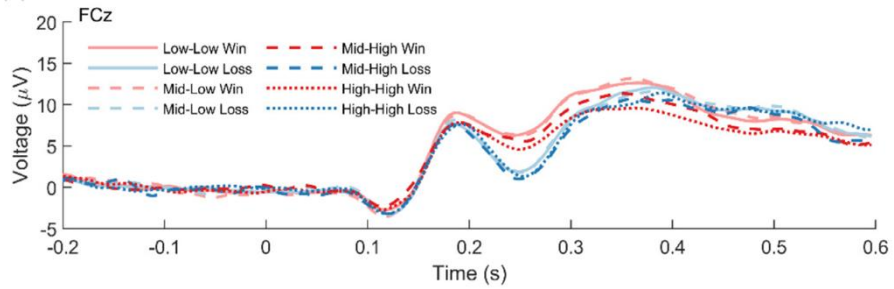
Milestone 5: Team „WHALE“

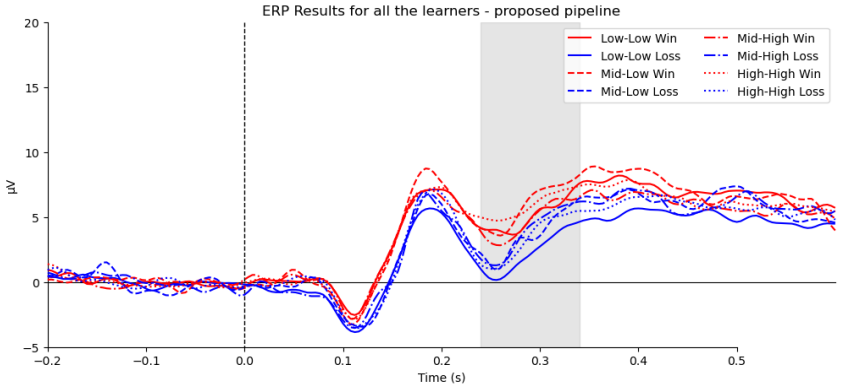
Members: Qianyue Li, Zheng Lin, Yanhong Xu **Dataset:** Reward Task

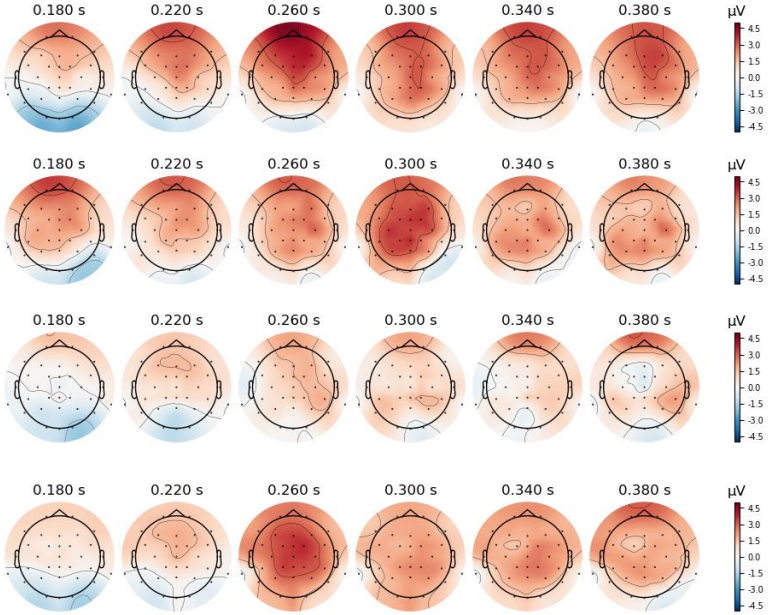
1. Main Research Question

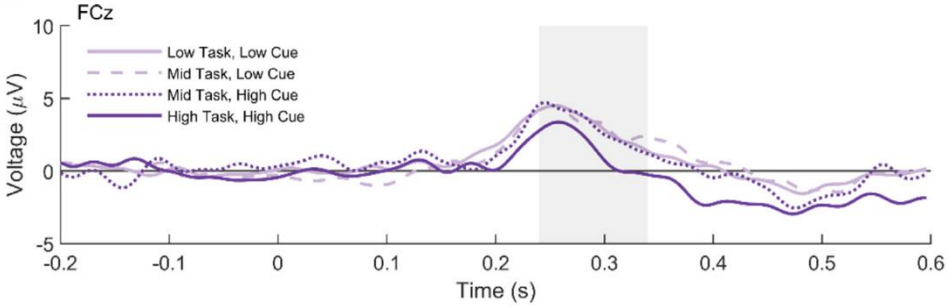
Whether and how reward-related anterior cingulate cortex (ACC) activity, as indexed by reward positivity (RewP) varies with task value.

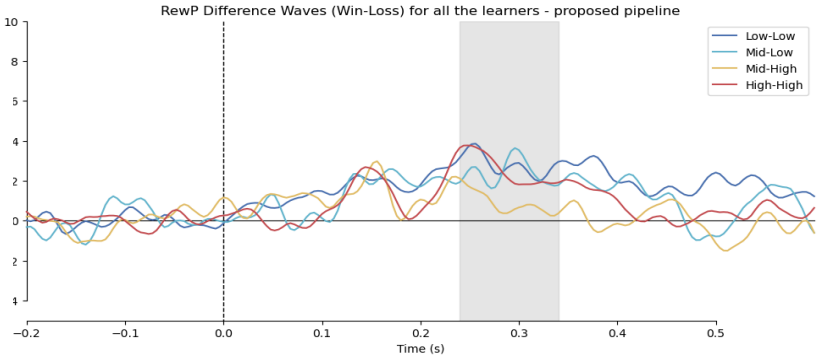
2. Feedback-locked EEG (RewP)











3. Repeated-measures t-tests for RewP

Learners, n=8

ML–LL:

- t(7)=0.043, p=.967, d=0.02; perm p=.993
- bootstrap CI_mean[−4.11, 3.23] → **crosses 0**

MH–HH:

- t(7)=−0.874, p=.411, d=−0.31; perm p=.354
- bootstrap CI_mean[−3.08, 1.01] → **crosses 0**

All participants, n=12 (sensitivity analysis)

ML-LL:

- t(11)=−0.343, p=.738, d=−0.10; perm p=.741
- bootstrap CI_mean[−3.20, 2.07] → **crosses 0**

MH-HH:

- t(11)=−2.442, p=.0327, d=−0.71; perm p=.0341
- bootstrap CI_mean[−4.22, −0.58] → **does not cross 0**

1. In learners (n=8), neither contrast is supported;
2. In sensitivity analysis (n=12), **MH–HH shows a negative effect** (perm p≈.034; CI excludes 0)
3. Needs confirmation with full data.

4. Behavioural results

1. The proportion of winning trials among different tasks (n=12):
   - low-value: 48.60%, 95% CI [46.71, 50.50]
   - mid-value: 59.03%, 95% CI [55.78, 62.28]
   - high-value: 60.00%, 95% CI [54.08, 65.92]
2. Statstical tests:
   - RM ANOVA: F(2,22) = 14.08, p = 0.0001155
   - partial eta^2 = 0.5614
   - generalized eta^2 = 0.4168

5. Decoding

- **Goal:** Decode **Win vs Loss** from EEG.
- **Data:** Use feedback-locked epochs and split into win and loss
- **Classifier:** linear logistic regression & SVM.
- **Metric: AUC** (0.5 = chance; >0.5 = discriminative), robust to imbalance.
- **Interpretation:** Above-chance AUC (+ permutation test) ⇒ **systematic outcome-related signal**, not just noise.